

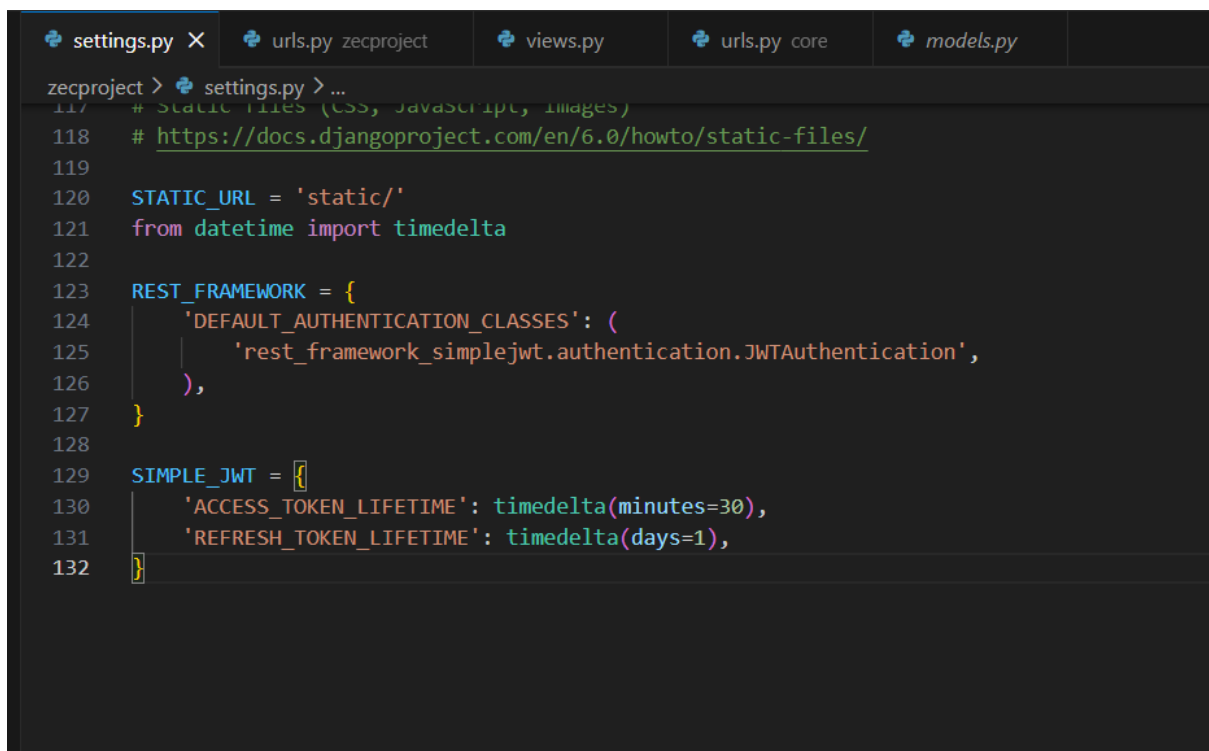
JWT Authentication & API Security

1. JWT Concepts

- Learned about JSON Web Tokens (JWT).
- Understood Access Token and Refresh Token.
- Studied stateless authentication.

2. JWT Configuration

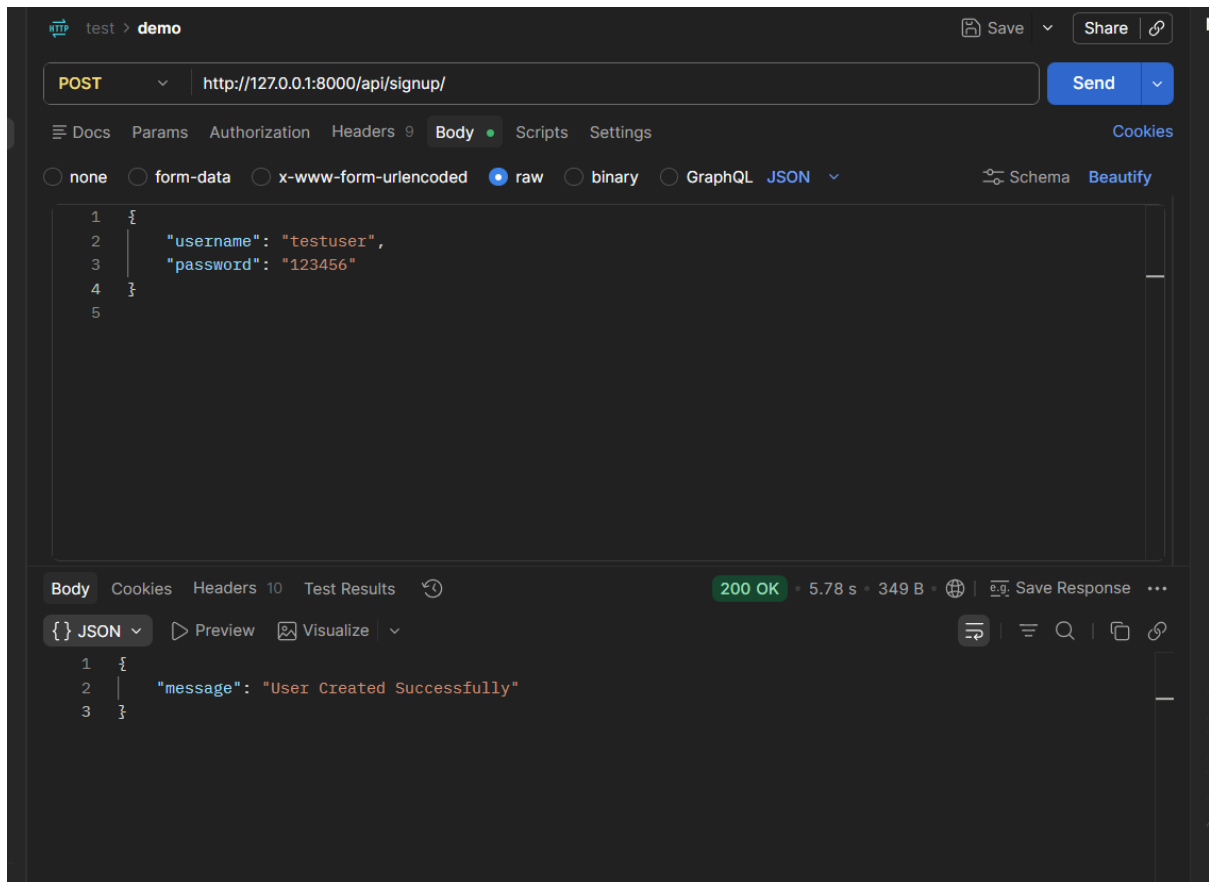
- Installed Django REST Framework SimpleJWT.
- Configured JWT authentication in Django settings.

A screenshot of a code editor with a dark theme. The editor has several tabs at the top: 'settings.py' (active), 'urls.py zecproject', 'views.py', 'urls.py core', and 'models.py'. The main window shows the 'settings.py' file with the following code:

```
117 # Static files (CSS, JavaScript, Images)
118 # https://docs.djangoproject.com/en/6.0/howto/static-files/
119
120 STATIC_URL = 'static/'
121 from datetime import timedelta
122
123 REST_FRAMEWORK = {
124     'DEFAULT_AUTHENTICATION_CLASSES': (
125         'rest_framework_simplejwt.authentication.JWTAuthentication',
126     ),
127 }
128
129 SIMPLE_JWT = {
130     'ACCESS_TOKEN_LIFETIME': timedelta(minutes=30),
131     'REFRESH_TOKEN_LIFETIME': timedelta(days=1),
132 }
```

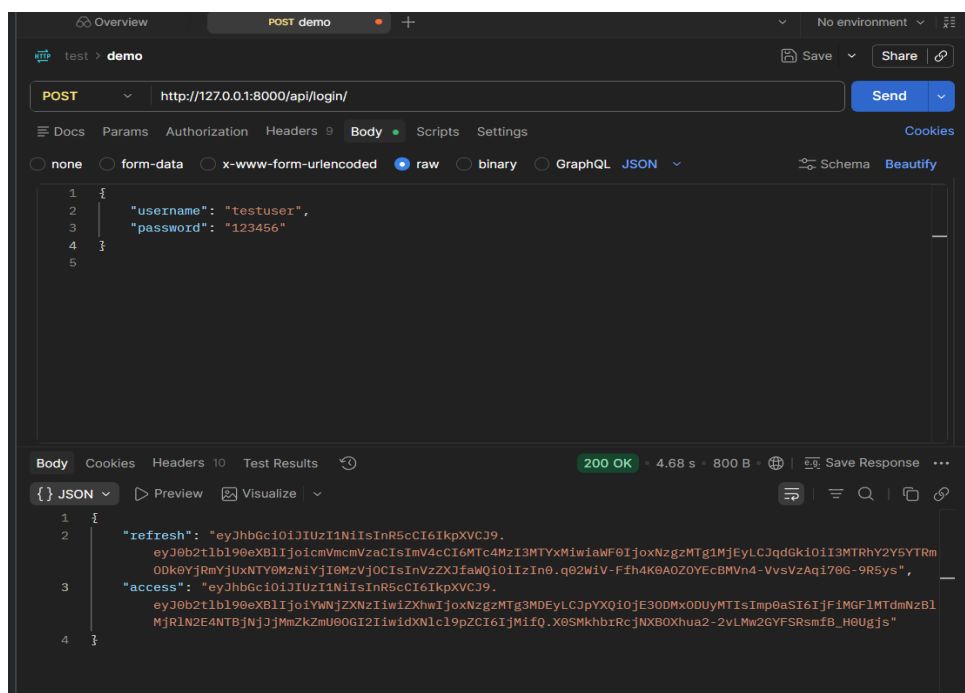
3. Signup API Development

- Created a Signup API.
- Registered new users through API requests.
- Verified successful user creation.



4. Login API Development

- Implemented JWT Login API.
- Generated Access Token and Refresh Token.
- Verified authentication using user credentials.



5. JWT Token Flow

- User registers using Signup API.
- User logs in using Login API.
- System generates Access Token and Refresh Token.
- Access Token is used for authenticated requests.

6. Protected Endpoint

- Created a protected API endpoint.
- Applied JWT authentication.
- Restricted access to authenticated users only.

